

Slide 1 — Title and Current Status

Slide title

Progress Presentation: Airflow Optimization for Thermal Comfort in Semi-Open Food Stalls

Content

- Project: Venturi-board airflow system
- Current stage: **pre-fabrication / simulation completion stage**
- Main update:
 - albedo paint removed due to budget
 - materials purchased
 - bengkel fabrication arranged
 - simulation almost completed
 - measurement and vendor feedback pending

Suggested status bar

Stage	Status
Proposal	Completed
Scope revision	Completed
Material purchase	Completed
Simulation	Almost completed
Prototype fabrication	Pending
Installation	Pending
Measurement	Pending
Vendor feedback	Pending

What to say

"This progress presentation focuses on what has changed since the proposal and what has been completed so far. Our project has moved from concept stage to material preparation and simulation validation stage."

Slide 2 — Original Plan vs Revised Plan

Purpose

Explain why the albedo paint was removed.

Item	Proposal Stage	Current Progress Stage
Albedo paint	Included	Removed
Venturi board	Included	Main focus
Main mechanism	Radiation + airflow	Airflow + convective/evaporative cooling
Main measurement	Temperature + wind speed	Wind speed before/after + thermal comfort
Budget	RM250	Focused on prototype materials and measurement

Key sentence

“Due to budget limitation, we removed the albedo paint and focused on a low-cost Venturi board that can be fabricated, installed, and tested within our available resources.”

Slide 3 — Completed Progress So Far

Purpose

Show real progress.

Content

Progress Item	Evidence	Status
Materials purchased	Photo / receipt / list	Completed
Anemometer purchased	Photo	Completed
Bengkel booking	Booking confirmation / verbal approval	Completed or in progress
Simulation model	Screenshot / graph	Almost completed
Prototype design	Sketch / CAD / schematic	Completed
Installation site	Vendor / stall location	Confirmed or pending

What to show visually

Use actual photos:

- bottles / board / frame material,
- anemometer,
- department bengkel,
- simulation screenshot,
- prototype sketch.

What to say

"At this stage, our major progress is that the project is no longer only conceptual. We have purchased the required materials, arranged fabrication support from the department workshop, and developed the simulation model."

Slide 4 — Current Prototype Design

Purpose

Show what you are going to build.

Include

- diagram / schematic,
- airflow direction,
- bottle funnel arrangement,
- frame structure,
- where it will be installed.

Physics labels

- inlet area (A_1)
- outlet area (A_2)
- incoming velocity (v_1)
- outlet velocity (v_2)
- user/stall region

Suggested wording

"The prototype uses recycled plastic bottles as converging ducts. Air enters through the wider opening and exits through the smaller neck, increasing local airflow velocity."

Slide 5 — Physics Laws Used

Purpose

Directly satisfy the rubric.

Use these four physics ideas.

1. Continuity equation

$$[A_1 v_1 = A_2 v_2]$$

Meaning: smaller outlet area gives higher outlet velocity under ideal flow.

2. Bernoulli principle

[$P + \frac{1}{2}\rho v^2 = \text{constant}$]

Meaning: increase in velocity is associated with pressure change.

3. Convective cooling

[$q_{\text{conv}} = hA(T_{\text{skin}} - T_{\text{air}})$]

Meaning: higher airflow increases (h), so heat leaves the body faster.

4. Humidity limitation

[$RH \uparrow \rightarrow \text{evaporation} \downarrow \rightarrow \text{cooling effectiveness} \downarrow$]

Meaning: in humid Malaysia, airflow may help comfort even if temperature does not drop much.

Strong sentence

“Our revised project does not mainly cool the air itself. It aims to increase local airflow, which improves convective and evaporative heat loss from the human body.”

Slide 6 — Simulation Progress

Purpose

Show that your simulation is nearly done and useful.

Content

- simulation objective,
- input variables,
- output variables,
- preliminary result / screenshot.

Suggested table

Simulation Component	Description
Input	inlet/outlet area, wind speed, temperature, humidity
Physics	continuity equation, comfort model
Output	predicted velocity increase, perceived cooling trend
Purpose	guide prototype design and measurement expectation

If you have a graph

Use one graph only, such as:

- humidity vs apparent comfort,
- inlet/outlet ratio vs velocity increase,
- before/after airflow prediction.

What to say

“The simulation is used to estimate how geometry and humidity affect performance before we build the prototype. It helps us predict what measurement range we should expect during field testing.”

Slide 7 — Measurement Plan

Purpose

Show exactly how you will prove the prototype works.

Before-after method

Measurement	Before Installation	After Installation
Air velocity	Measure at same point without board	Measure at same point with board
Temperature	Record ambient temperature	Record ambient temperature
Humidity	Record RH if available	Record RH if available
User comfort	Ask vendor/customer	Ask vendor/customer

Measurement setup

- Use anemometer bought by group
- Measure at:
 - front of board,
 - outlet of board,
 - working/customer zone
- Repeat readings:
 - 3–5 times per point
- Use average value
- Keep same time period if possible

Suggested formula

[% \text{ improvement} = \frac{v_{\text{after}} - v_{\text{before}}}{v_{\text{before}}} \times 100\%]

What to say

“Our main quantitative evidence will be the percentage increase in local air velocity before and after installing the Venturi board.”

Slide 8 — Community Feedback Plan / Preliminary Feedback

Purpose

Handle the feedback rubric honestly.

Since final vendor feedback is not collected yet, split it into:

1. preliminary feedback, if any,
2. planned post-installation feedback.

Preliminary feedback example

If vendor already commented:

Feedback	Meaning	Planned Response
Prototype appearance may look messy	Adoption issue	Cover frame with cloth or cleaner casing
Installation should not disturb business	Practical constraint	Use removable mounting
Airflow should face customer/worker area	Direction issue	Adjust orientation

Planned direct quotes

You cannot fake final quotes. But you can prepare the structure:

- Vendor quote after seeing prototype
- Worker quote after airflow test
- Customer or nearby user quote after installation

Feedback form questions

Question	Scale
Airflow is noticeable	1–5
The stall feels more comfortable	1–5
Prototype appearance is acceptable	1–5

Question	Scale
Installation does not disturb operation	1–5
Would you keep this prototype?	Yes/No

Negative comment

Your expected or preliminary negative comment:

“The plastic bottle prototype may look messy and affect the image of the stall.”

Revision

- cover with cloth,
- use cleaner frame,
- keep bottle openings unobstructed.

Slide 9 — Current Challenges and Physics-Based Revisions

Purpose

Show maturity and satisfy “what changed after feedback.”

Challenge / Feedback	Physics or Practical Meaning	Revision
Albedo paint too expensive	Cannot reduce solar heat input	Focus on airflow comfort
Natural wind may be weak	Venturi needs inlet airflow	Place near fan or wind-facing side
Humidity may reduce cooling	Evaporation less effective	Include humidity in simulation
Prototype looks unattractive	Vendor may reject installation	Add cloth/casing
Measurements may fluctuate	Outdoor airflow is unstable	Repeat readings and average

Strong sentence

“The feedback and constraints did not change the physical laws, but they changed our boundary conditions, design priorities, and success metrics.”

Slide 10 — Remaining Work

Purpose

Show what still needs to be completed.

Clear remaining checklist

Remaining Task	Purpose	Status
Build prototype at bengkel	Convert design into hardware	Pending
Install at stall	Field implementation	Pending
Measure before/after airflow	Quantitative validation	Pending
Measure temperature/humidity	Environmental context	Pending
Collect vendor feedback	Community validation	Pending
Interpret data physically	Final analysis	Pending

What to say

"The remaining work is mainly experimental validation. Once the prototype is built and installed, we will compare airflow before and after, then connect the measurement results to the physics model."

Slide 11 — Updated Timeline and Contingency Plan

Updated timeline

Week	Task
Current week	Finalize simulation and bengkel booking
Next week	Build prototype
Following week	Install prototype and collect before/after data
After measurement	Analyze data and collect feedback
Final stage	Improve design and prepare final report

Contingency plan

Risk	Backup Plan
Bengkel unavailable	Use simpler hand tools / reschedule early
Natural wind too weak	Test near fan or during windier period
Vendor rejects appearance	Add cloth/casing
Sensor readings unstable	Repeat measurement and average
Installation not allowed	Use temporary non-invasive mounting
Simulation differs from data	Explain real-world losses: turbulence, leakage, wind direction

Slide 12 — Conclusion

Purpose

End with a confident but honest status update.

Suggested conclusion

"Our project has progressed from proposal to implementation preparation. The albedo paint component was removed due to budget constraints, so the project now focuses on a Venturi-board airflow system. We have purchased the materials, arranged access to the department workshop, and nearly completed the simulation model. The next stage is to fabricate and install the prototype, measure before-and-after airflow using the anemometer, and collect vendor feedback. The final interpretation will compare experimental results with the physics model and explain the effect of airflow and humidity on perceived thermal comfort."